25901441 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY202 - EXPLORING THE COSMOS: OBSERVATIONS, CELESTIAL BODIES, AND COSMIC EVOLUTION

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. What defines a Zodiac constellation? [U] / [CO1]
2. What is the primary cause of Earth's seasons? [U] / [CO1]
3. Write the electromagnetic spectrum in the increasing order of frequency. [U] / [CO2]
4. What happens to the parallax angle as the distance to a star increases? [K] / [CO2]
5. Define focal ratio of a telescope and explain its physical significance. [U] / [CO3]
6. What is a neutron star? [U] / [CO4]
7. Write the net reaction of the p–p chain. [U] / [CO4]
8. Explain how the surface temperature of stars is related to colour. [An] / [CO4]
9. What are the possible final stages of most massive stars? [An] / [CO4]
10. Why the massive stars evolve rapidly and have less main sequence life time? [An] / [CO4]

11. What is an absorption spectrum? [U] / [CO4]

12. Draw the light curve of a typical Cepheid variable star. [U] / [CO5]

13. What is cosmic microwave background radiation (CMBR)? [U] / [CO5]

14. What is Big Bang nucleosynthesis? [U] / [CO5]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Give an account on the Equatorial coordinate system. In what scenarios is it more useful than the Ecliptic coordinate system? [U] / [CO1]

16. Distinguish between flux and luminosity. If a star has a luminosity of 3×10^{26} W and is located at a distance of 10 pc, calculate the flux received at Earth. [A] / [CO2]

17. Write a note on the antennas used in radio telescopes. [U] / [CO3]

18. Describe the concept of event horizon in detail. [U] / [CO4]

19. Explain any theoretical method to calculate the temperature of a star. [U] / [CO4]

20. Compare spiral and elliptical galaxies using observable properties such as star population, gas content, and structure. [A] / [CO4]

21. The Hα spectral line (emitted wavelength = 656.3 nm) from a galaxy is observed at 670.2 nm. Estimate the redshift and recessional velocity of the galaxy. [A] / [CO5]

22. The apparent magnitude of the galaxy NGC3159 is 14.4, and recessional velocity with respect to the Milky Way 6940 km/s. Find the distance and absolute magnitude of the galaxy. (Given Ho = 70 km/s/Mpc) [A] / [CO5]

[5x6 = 30]